

ZeroPad3d#

[https://pytorch.org/docs/stable/generated/torch.nn.modules.padding.ZeroPad3d](https://pytorch.org/docs/stable/generated/torch.nn.modules.padding.ZeroPad3d.html)

Description:

Pads the input tensor boundaries with zero. For N-dimensional padding, use `torch.nn.functional.pad()`. `padding` (int, tuple) – the size of the padding. If is int, uses the same padding in all boundaries. If a 6-tuple, uses `(padding_left\text{padding\_left}padding_left, padding_right\text{padding\_right}padding_right, padding_top\text{padding\_top}padding_top, padding_bottom\text{padding\_bottom}padding_bottom, padding_front\text{padding\_front}padding_front, padding_back\text{padding\_back}padding_back)` Input: $(N, C, D_{in}, H_{in}, W_{in})$ or $(N, C, D_{in}, H_{in}, W_{in})$ or $(C, D_{in}, H_{in}, W_{in})$. Output: $(N, C, D_{out}, H_{out}, W_{out})$ or $(N, C, D_{out}, H_{out}, W_{out})$ or $(C, D_{out}, H_{out}, W_{out})$, where

Function Signature

```
class torch.nn.modules.padding.ZeroPad3d(padding)[source]#
```

Parameters

Shape:

```
extra_repr()[source]#
```

Return type

Returns:

str

Code Examples

```
>>> m = nn.ZeroPad3d(3)
>>> input = torch.randn(16, 3, 10, 20, 30)
>>> output = m(input)
>>> # using different paddings for different sides
>>> m = nn.ZeroPad3d((3, 3, 6, 6, 0, 1))
>>> output = m(input)
```

Detailed Documentation

Documentation

Pads the input tensor boundaries with zero.

For N-dimensional padding, use `torch.nn.functional.pad()`.

padding (int, tuple) – the size of the padding. If is int, uses the same padding in all boundaries. If a 6-tuple, uses (`padding_left\text{padding\_left}`padding_left, `padding_right\text{padding\_right}`padding_right, `padding_top\text{padding\_top}`padding_top, `padding_bottom\text{padding\_bottom}`padding_bottom, `padding_front\text{padding\_front}`padding_front, `padding_back\text{padding\_back}`padding_back)

Input: (N,C,Din,Hin,Win)(N, C, D_{in}, H_{in}, W_{in})(N,C,Din,Hin,Win) or (C,Din,Hin,Win)(C, D_{in}, H_{in}, W_{in})(C,Din,Hin,Win).

Output: (N,C,Dout,Hout,Wout)(N, C, D_{out}, H_{out}, W_{out})(N,C,Dout,Hout,Wout) or (C,Dout,Hout,Wout)(C, D_{out}, H_{out}, W_{out})(C,Dout,Hout,Wout), where

$D_{out} = D_{in} + \text{padding_front} + \text{padding_back}$
 $D_{out} = D_{in} + \text{padding_front} + \text{padding_back}$

$H_{out} = H_{in} + \text{padding_top} + \text{padding_bottom}$
 $H_{out} = H_{in} + \text{padding_top} + \text{padding_bottom}$

$W_{out} = W_{in} + \text{padding_left} + \text{padding_right}$
 $W_{out} = W_{in} + \text{padding_left} + \text{padding_right}$

Return the extra representation of the module.
